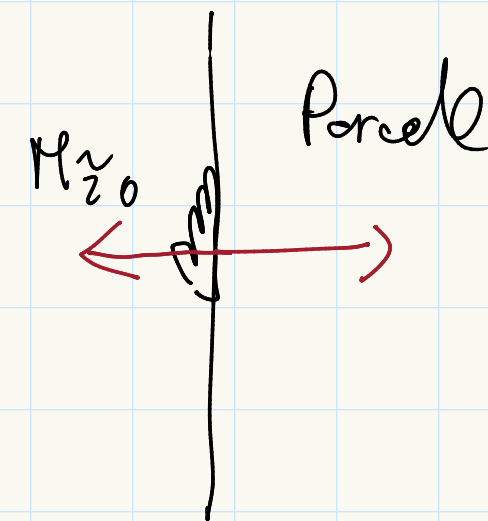
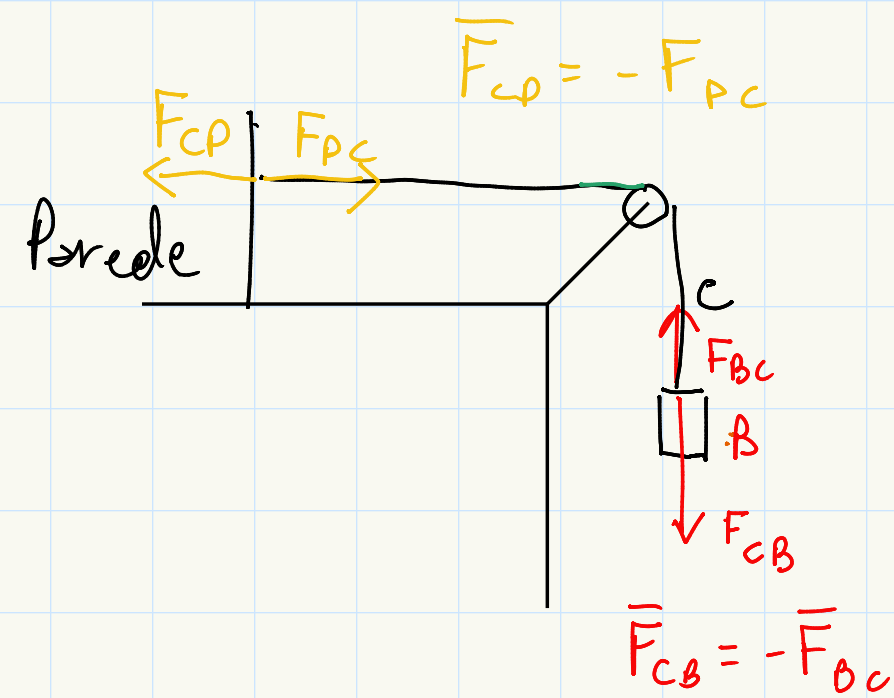
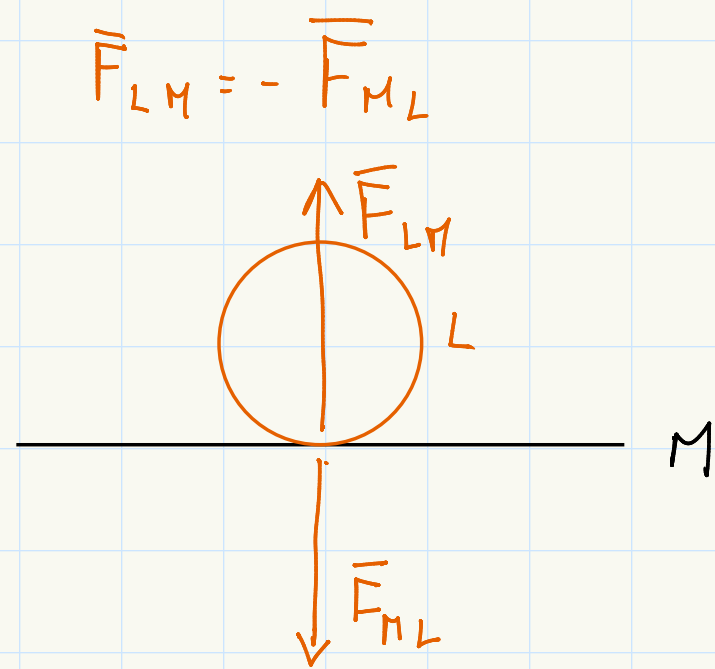
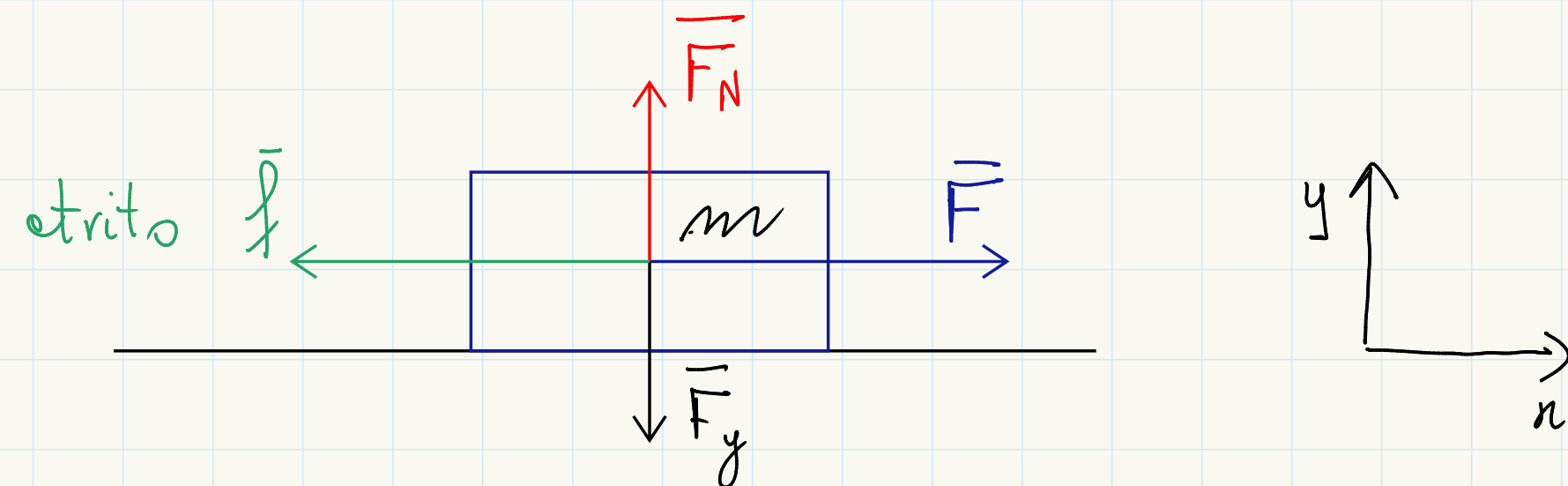




Atrito



$$\vec{F}_g = m \vec{g} \vec{j}$$

$$m\vec{a} = \vec{F} + \vec{f}, \quad \vec{f} = -\mu \vec{F}_N$$

μ coeficiente de atrito $\begin{cases} \mu_s, \text{estático} \\ \mu_d, \text{dinâmico} \end{cases}$

$$x: m a_x = F - \mu F_N$$

$$y: m a_y = F_N - m g, \quad F_N = m g \leadsto \text{se o plano é horizontal } \vec{F}_N = m g \vec{j}$$

$$\leadsto m a_x = F - \mu m g$$